DO NOW — RECALL

OOP: testing and debugging

Name: _____ Class: _____ Date: _____

1. Insert 12 into a BST node 10 with empty right child. Where does it go?

2. What is a reference?

3. What is a local variable?

4. If an object's Balance=20 and Deposit(5) correctly adds five, what is Balance?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

As the right child of 10.

2. Model answer

A value that identifies an object rather than copying the object itself.

3. Model answer

A variable declared inside a method or block with limited scope.

4. Model answer

25.

Vocabulary — know these terms

encapsulation — Hide implementation details and control state through methods/properties.

constructor — Initialises a new object.

field / property — Stores or exposes object state.

breakpoint — Pauses execution at a selected code line.

boundary test — Checks values at or just beyond a permitted limit.

NEW LEARNING

Read, annotate and explain

Inspect class state

The object has a Balance property initially set by its constructor.

Trace method inputs

Deposit(5) passes amount=5. Since amount>0, Balance+=5 executes.

Debug the outcome

Expected Balance=25. If unchanged, check whether the code assigns to Balance or merely calculates a temporary value.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find and correct the fault when AddPoints(3) fails to update Score=7.

Predict first

The local newScore becomes 10, but the field Score remains 7.

Model the reasoning

Replace the local-only calculation with Score += points;.

Model the reasoning

Retest: after AddPoints(3), Score should equal 10.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

An object starts at Count=4 and runs Increment() which performs Count += 1. What is Count after three calls?

Question 2

Correct `public void SetAge(int age) { int Age = age; }` where Age is an existing property. Explain why it fails.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A BankAccount.Withdraw(amount) rejects only amounts <=0, but permits overdrafts. Add validation and give boundary tests for Balance=50.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A method may calculate a correct new value without actually updating the object's property.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.